
DREAMER SYSTEMS

THREE DESIGN APPROACHES

Integrated Cube | Adaptive Airframe | Scalable 3-Chamber Platform

A curated concept portfolio preserving the original engineering pages and visual studies.

CONCEPT PORTFOLIO | JULY 2026

Three distinct answers to the same cultivation challenge

1

INTEGRATED CUBE

PRODUCT-VISION APPROACH

- Premium autonomous single-cabinet concept
- Solar and battery integrated from the start
- Best for a polished finished-system vision
- Phase 1 validates climate hardware before cabinetry

2

ENGINEERING-FIRST APPROACH

- One chamber organized around a service spine
- Pressure-balanced floor and removable cassette
- Best for airflow, moisture and failure-mode testing
- Most rigorous path for proving mechanisms

3

SCALABLE OPERATIONS APPROACH

- Three isolated chambers share utilities
- Balanced manifold, low-voltage power and sensors
- Best for modular throughput and operational scaling
- Replicate only after one chamber passes validation

The chapters retain the original source pages at native dimensions.

CHAPTER 01

INTEGRATED DREAMER CUBE



A premium autonomous chamber presented as a finished product vision, supported by a buildable Phase 1 climate prototype.

WHAT THIS APPROACH OPTIMIZES

- System integration
- Off-grid operation
- Product-level enclosure
- User-facing controls
- Prototype-first risk reduction

DREAMER CUBE

SOLAR-POWERED AUTONOMOUS MYCOCHAMBER

CONTROLLED ENVIRONMENT. MAXIMUM YIELD.

An all-in-one outdoor cultivation system that combines advanced environmental control, intelligent automation, and renewable energy for year-round, high-performance mushroom production.

KEY INNOVATIONS

- Energy Independent**
Solar + Battery for true off-grid operation
- Modular Grow Cassettes**
Pull-out trays for easy handling and cleaning
- Intelligent Climate Control**
AI-driven environment for optimal fungal growth
- Designed for Simplicity**
Serviceable components + clean, tool-less access
- Built for the Long Haul**
Marine-grade enclosure rated for extreme conditions

SYSTEM HIGHLIGHTS

- 3x GROW CASSETTES**
- 40-95% HUMIDITY CONTROL**
- 60-80°F TEMPERATURE RANGE**
- <35 dBA WINDSPEED QUIET OPERATION**
- SOLAR + BATTERY 24/7 AUTONOMOUS OPERATION**
- IP55 WEATHER RESISTANT**

01 SYSTEM OVERVIEW



AIRFLOW CONCEPT

- Fresh Air In (Filtered)
- Conditioned Air Flow
- Return Air Flow
- Exhaust Air Out

IDEAL FOR

- Home growers
- Off-grid locations
- Commercial micro-farms
- Research & education

02 MECHANICAL ARCHITECTURE

- INSULATED GASKET LID**
Marine-grade seal with compression latches for air-tight closure.
- REMOVABLE GROW CASSETTE**
Slide-out cassette with adjustable rails. Holds standard substrate pans or custom trays.
- PERFORATED AIR PLENUM**
Distributes conditioned air evenly across all cassettes.
- AIR DISTRIBUTION MANIFOLD**
Precision manifold delivers balanced airflow to each cassette level.
- DRAINAGE SYSTEM**
Sloped floor channels water to rear drain. Includes trap and quick-drain port.
- UTILITY SPIKE**
Central spine houses all mechanical, electrical, and plumbing components.
- SERVICE ACCESS PANEL**
Tool-less panel for electronics and system maintenance.
- BASE ISOLATION FEET**
Adjustable, anti-vibration feet for stable positioning on any surface.



EXTERIOR DIMENSIONS



CASSETTE DIMENSIONS (INSIDE)



SYSTEM WEIGHT (EST.)

- Empty: 95 lb (43 kg)
- Operating: 145 lb (66 kg)

03 POWER & CONTROL SYSTEM

SOLAR POWER SYSTEM

- 400W SOLAR PANELS (x2)**
High-efficiency panels with aluminum mounting frame.
- MPPT CHARGE CONTROLLER**
Victtron SmartSolar MPPT 250/70 for maximum harvest.
- DC DISCONNECT**
Weatherproof safety switch for maintenance and isolation.

BATTERY SYSTEM

- BLUETTI AC70 BATTERY**
768Wh LiFePO4 battery provides stable, long-life power.
- BATTERY HEATER PAD**
Maintains optimal temperature for battery performance.
- BATTERY VENTILATION**
Active intake & exhaust prevents overheating.



CONTROL & ELECTRONICS

- ESP32 CONTROLLER**
Dual-core processor running Dreamer OS.
- SENSOR HUB**
Collects data from all sensors and manages system I/O.
- RELAY BOARD**
Controls high-power devices (switching, fans, heater, etc.).
- USB SERVICE PORT**
Local access for updates and diagnostics.
- STATUS LIGHT BAR**
System status at a glance.

POWER BUDGET (TYPICAL)	
Component	Power (W)
Circulation Fans	8
Humidifier	12
Electronics	3
Sensors	1
Heater (avg)	10
Total Average	34 W
Peak (heater on)	75 W

BATTERY RUNTIME (AC70)

- Average Load (34W): ~22.5 hrs
- Peak Load (75W): ~10.5 hrs
- (At 100% battery capacity)

* Actual runtime varies with settings and environment.

AC OUTPUT (OPTIONAL)



04 SYSTEM INTEGRATION & DREAMER OS

DREAMER OS DASHBOARD



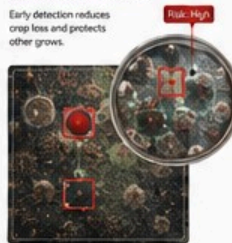
AI ENVIRONMENT CONTROL

- AI continuously learns and adjusts setpoints to maximize growth while minimizing energy usage.
- Auto setpoint tuning
- VPD optimization
- Adaptive airflow
- Predictive humidification

HARVEST PREDICTION



CONTAMINATION DETECTION



SYSTEM INTEGRATION DIAGRAM



BUILT FOR RELIABILITY

- IP55 WEATHER RESISTANT
- MARINE GRADE ALUMINUM
- CORROSION RESISTANT
- OPERATES IN EXTREME -4°F TO 122°F
- DESIGNED FOR 10+ YEARS

SCALABLE ECOSYSTEM

Connect multiple Dreamer Cubes and manage from a single dashboard.



DREAMER CUBE

PHASE 1 BUILD GUIDE

One battery-powered, solar-ready autonomous fruiting chamber

Prototype objective: prove the climate system before building the finished cabinet

DESIGN BASIS

CHAMBER	POWER	CHARGING	CONTROL
80-qt gasket tote	Bluetti AC70	200W solar	Inkbird + sensors

Phase 1 finished result

- One 80-quart gasketed tub in front-opening sideways orientation
- One removable grow tray or cassette with a raised internal shelf
- One low-speed circulation fan plus passive fresh-air openings
- External ultrasonic humidifier controlled by an Inkbird IHC-200
- Bluetti AC70 battery power with solar charging capability
- Basic temperature, humidity and energy logging
- Measured battery runtime and daily energy use before scaling

Success criteria

☐	Stable humidity without direct fog striking the substrate
☐	Gentle air movement without excessive surface drying
☐	Controlled condensate and dry electrical connections
☐	Overnight battery coverage and measurable solar recovery
☐	Automatic restart after power interruption
☐	At least 24 hours of safe unattended operation
☐	Easy cleaning, tray removal and component access

1. Choose the test location

Use a garage, basement, utility room or sheltered outdoor work area for initial testing.

- Allow roughly 36 in wide x 30 in deep x 48 in high.
- Provide access to sunlight for the solar panel.
- Keep a dry, shaded location available for the AC70.
- Avoid direct vehicle exhaust and rain exposure.
- Keep the battery outside the humidified chamber.

2. Buy the Phase 1 hardware

Purchase only the components needed to prove one complete chamber.

Subsystem	Required components
Chamber	80-qt gasket-lid tote; wire shelf; shallow grow tray; drip tray; stainless hardware; neutral-cure silicone; PTFE tape
Airflow	Adjustable low-speed fan; fan guard; coarse washable filter; filter patches or micropore tape
Humidity	Ultrasonic humidifier with auto-restart; Inkbird IHC-200; silicone hose; bulkhead/barb; clamps
Power	Bluetti AC70; 200W panel; compatible solar cable; grounded power strip; plug-in watt meter
Monitoring	Independent digital temp/RH sensor, or ESP32 plus quality temp/RH sensor and USB cable

3. Bench-test every electrical component

Test the equipment before drilling or modifying the tub.

- Fully charge the AC70.
- Connect and test the fan at its lowest setting.
- Plug the humidifier into the Inkbird humidity-controlled outlet.
- Confirm the humidifier resumes automatically after power is restored.
- Measure each component individually with the watt meter.

BUILD NOTE: Do not drill the tub until the humidifier passes the automatic-restart test.

4. Rotate and support the tub

Set the tote on its side so the gasket lid opens toward you.

- Support it on a flat shelf or two rails.
- Confirm the lid opens fully and the gasket remains seated.
- Check the shell for bowing before adding load.
- Keep all mounting temporary until fit is confirmed.

5. Install the internal shelf

The shelf creates a raised grow zone above the air-distribution and drainage area.

- Position the shelf about 3-5 in above the new bottom.
- Mark four to six mounting points.
- Use the smallest practical holes and large load-spreading washers.
- Seal all penetrations with silicone and allow full cure.
- Load-test with more weight than the expected substrate.
- Verify the tray slides in and out without removing the shelf.

6. Create passive fresh-air openings

Begin conservatively because holes can be enlarged later.

- Use approximately eight filtered openings.
- Place roughly two on each side near substrate or tray height.
- Space openings instead of clustering them.
- Avoid placing all openings at the top.
- Tune later by changing filter density or partial coverage.

7. Install the circulation fan

The fan should circulate air, not blast the grow surface.

- Mount high on the rear or side wall.
- Aim airflow along a wall rather than at the substrate.
- Use a chamber-side guard and vibration-isolating washers.
- Use a sealed cable gland for the wire.
- Set the fan to the lowest practical speed.

BUILD NOTE: At tray height, airflow should feel gentle rather than drafty.

8. Install the humidity inlet

Introduce fog indirectly so it disperses before reaching the grow tray.

- Place the inlet low on one side or below the shelf.
- Direct the inlet toward a wall or under-shelf plenum.
- Install the bulkhead with gasket and sealant.
- Route the hose without loops, dips or low spots.
- Whenever practical, slope the hose back toward the humidifier.
- Secure the hose so it cannot pull on the tub wall.

9. Place the humidity probe

The probe should read the chamber environment rather than the fog plume.

- Mount near tray height and toward the center.
- Keep several inches away from the fog inlet.

- Avoid direct fan airflow and dripping surfaces.
- Use a removable mount for cleaning and calibration.

10. Add basic drainage

Keep drainage simple for the prototype.

- Start with a removable drip tray under the shelf.
- Optionally add a low-point drain and external bottle later.
- Inspect corners, hose entry, lid gasket and shelf mounts after several hours.
- Do not run unattended until water accumulation points are understood.

Power architecture

Solar panel -> AC70 solar input -> AC70 battery/inverter

AC70 AC outlet -> watt meter -> power strip -> Inkbird -> humidifier

AC70 USB/DC output -> compatible fan and ESP32/sensors

11. Assemble the power system

- Keep the battery outside the humidified chamber.
- Elevate AC connections and create drip loops in every cord.
- Keep electrical components away from the drain bottle.
- Protect solar cables from doors, sharp edges and standing water.
- Do not connect a fan to 12V until its input specification is verified.

12. Run a 24-hour empty-chamber test

Operate the system without substrate before beginning a live grow.

- Record humidity rise time and overshoot.
- Locate condensation and hose water traps.
- Check whether one area receives excessive airflow.
- Watch for direct fogging of the probe.
- Record battery percentage after 24 hours.
- Record solar energy recovered during daylight.

BUILD NOTE: Begin with a wider humidity differential to avoid rapid on/off cycling.

13. Measure the real energy budget

Record both instantaneous power and full-day energy use.

- Fan watts
- Humidifier watts while active
- Controller and sensor watts
- Total idle watts
- Total watts during humidification

- Humidifier run time over 24 hours
- Overnight battery percentage used
- Solar watt-hours produced

DAILY ENERGY USE = AVERAGE SYSTEM WATTS x 24 HOURS

Example: 20 W average x 24 h = 480 Wh/day

14. Tune airflow and humidity

Change only one variable at a time and log the result.

Observed condition	Adjustment sequence
Chamber too wet	Lower target; widen differential; increase passive exhaust slightly; reduce humidifier output; improve drainage
Surface drying	Lower fan speed; aim fan at wall; add diffuser; reduce nearest opening; move tray
Top-to-bottom variation	Redirect fog inlet; improve under-shelf distribution; slightly increase circulation; reposition probe

15. Perform simulated failure tests

Confirm the prototype fails safely before introducing a live grow.

- Disconnect solar and confirm battery operation continues.
- Turn off humidification and measure humidity decay.
- Turn off the fan and confirm passive airflow remains available.
- Test probe disconnection behavior safely.
- Simulate battery shutdown and restore power.
- Confirm every required component restarts automatically.

16. Run a one-week biological test

Use one manageable grow tray and observe the system twice daily.

- Temperature and RH
- Battery state and solar input
- Humidifier duty cycle
- Condensation and surface moisture
- Drying patterns and growth uniformity
- Contamination signs
- Daily maintenance time

BUILD NOTE: Keep AI control disabled during Phase 1. First learn what normal operation looks like.

17. Decide whether Phase 1 passed

Freeze the Phase 2 cabinet specifications only after the prototype meets all success criteria.

Recommended build schedule

Timing	Work package
Day 1	Finalize parts and bench-test electrical equipment
Day 2	Rotate the tub and install the internal shelf
Day 3	Drill passive openings and install the fan
Day 4	Install humidity plumbing and probe
Day 5	Assemble battery and solar power
Day 6	Run the empty 24-hour test
Days 7-10	Tune airflow, humidity and drainage
Days 11-17	Run one live grow tray and collect data
Day 18	Review results and freeze Phase 2 specifications

PHASE 1 RULE: PROVE THE CLIMATE SYSTEM FIRST. MAKE IT BEAUTIFUL SECOND.

AX-80 ADAPTIVE AIRFRAME

A hardware-first chamber architecture separating the biological payload from airflow, moisture, drainage and service mechanics.

WHAT THIS APPROACH OPTIMIZES

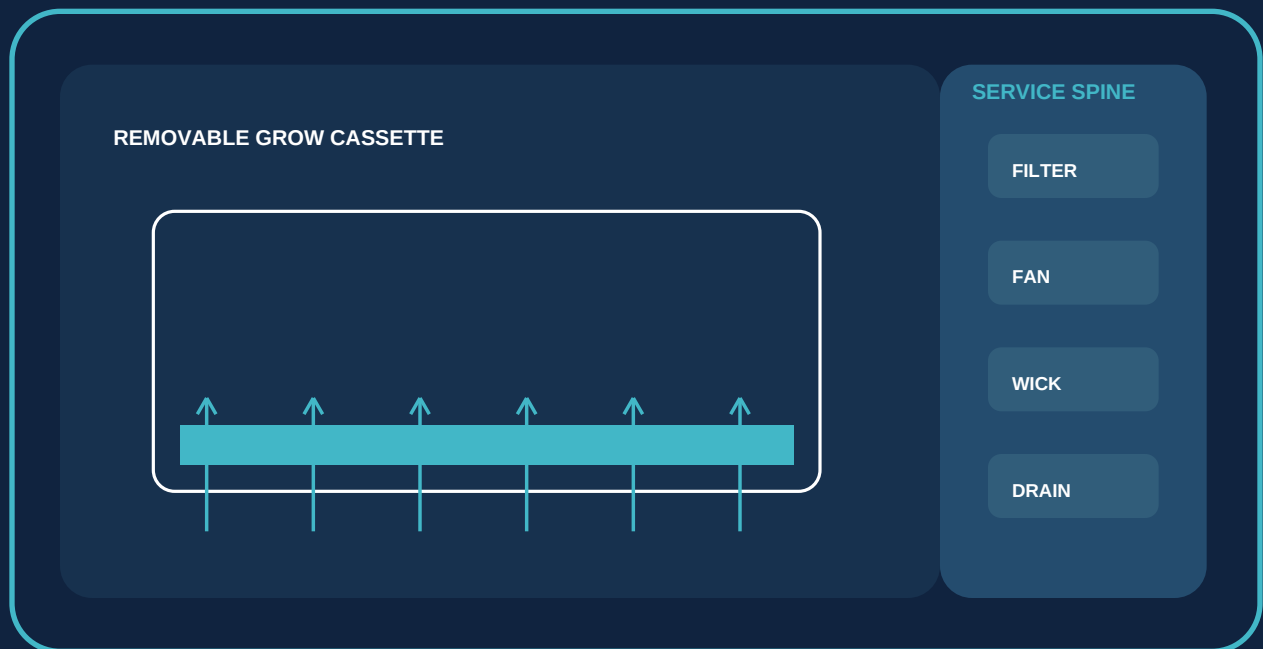
- Pressure-balanced airflow
- Condensate control
- Removable grow cassette
- External service access
- Graceful passive failure

DREAMER SYSTEMS

AX-80

Adaptive Fruiting Chamber

A hardware-first engineering brief for a pressure-balanced, serviceable and failure-tolerant 80-quart fruiting platform.



CONCEPT RELEASE 01

Hardware architecture, airflow rationale, build specification, and efficacy justification.

JULY 2026

DESIGN THESIS

Separate biological space from mechanical space.

The AX-80 conditions air in a sealed sidecar, distributes it through a low-pressure floor, and houses the crop in a removable cassette. The tote becomes a shell; the airframe becomes the machine.

3 zones

One integrated chassis

service / plenum / grow

0 jets

Direct fog exposure

diffused delivery only

2 modes

Powered + passive

graceful failure

THE SYSTEM IN ONE VIEW

1 SERVICE SPINE

Filter

Fan + humidity

Drain + bypass

2 FLOOR PLENUM

Pressure equalization

Tapered outlet array

Reversible airflow

3 GROW CASSETTE

Slide-in pan frame

Clean crop volume

Tool-free removal

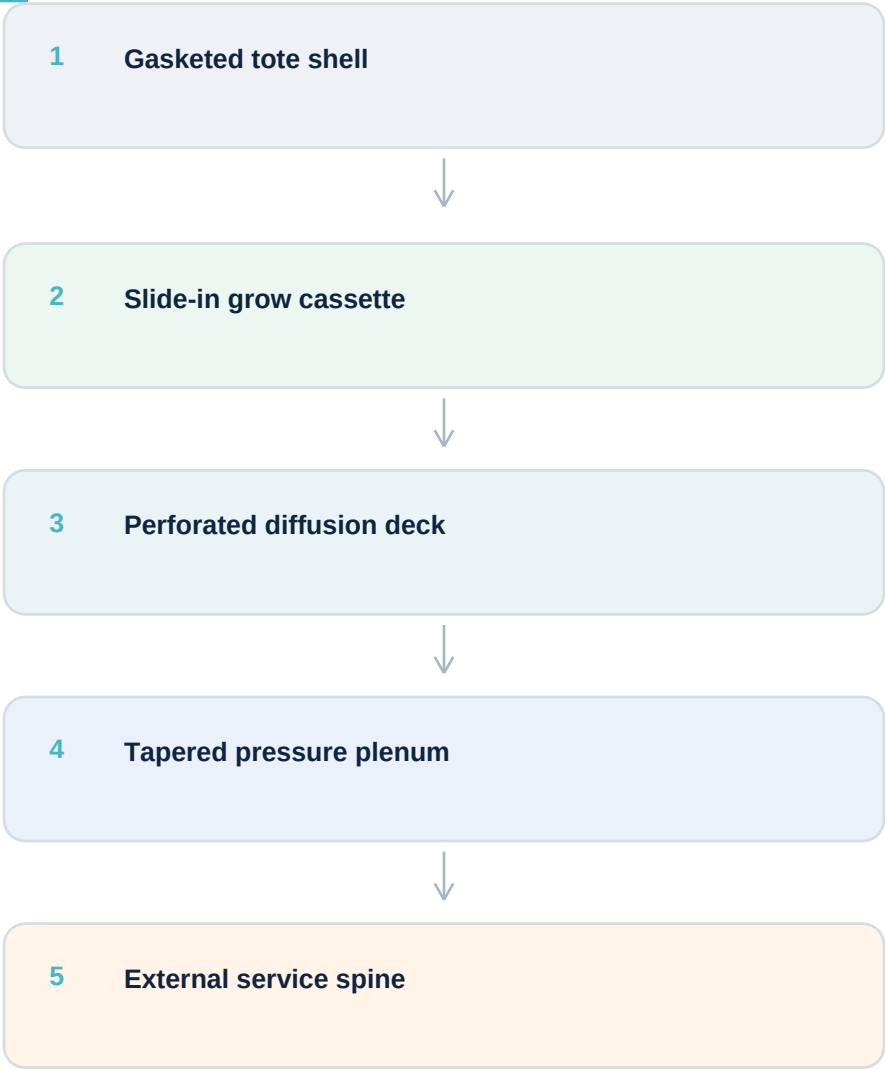
Central innovation

A reversible plenum permits upward displacement during normal operation and cross-sweep purge during recovery.

Practical innovation

All wet and powered components are reachable from one exterior service hatch without disturbing the crop.

EXPLODED SIDE ELEVATION



INTERFACES

Dry boundary

Sensors and crops remain outside the fog-generation path.

Wet boundary

Condensate is confined to the service spine and drain channel.

Power boundary

All electrical connectors remain external and drip-looped.

Cleaning boundary

Cassette, deck and plenum lift out independently.

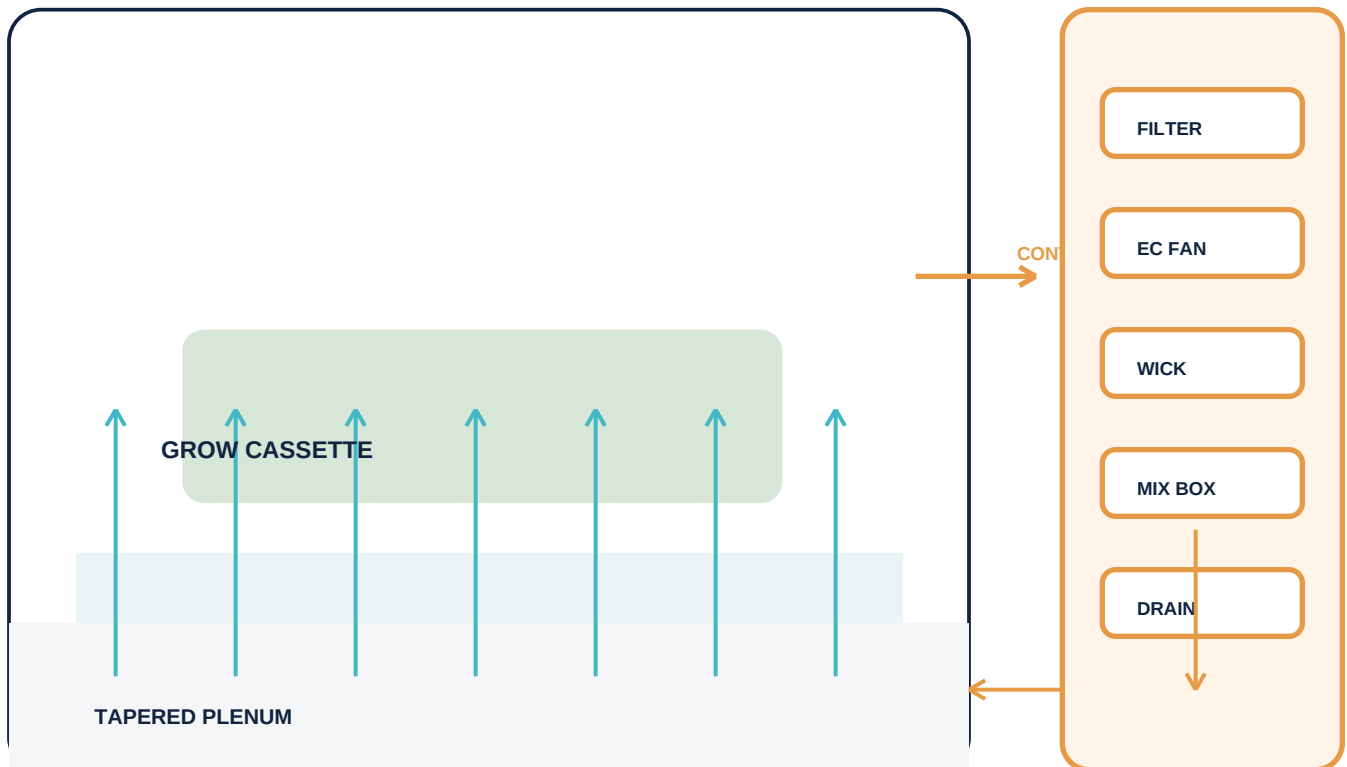
Upgrade boundary

Fan, filter, wick and control modules use standardized mounts.

RECOMMENDED ENVELOPE

External shell	80 qt gasket tote	approx. 32 x 19 x 17 in
Service spine	4.0-5.0 in deep	full rear width
Plenum height	1.75-2.25 in	below grow cassette
Cassette clearance	2.0 in minimum	around pan perimeter
Drain pitch	1-2 degrees	toward external trap

NORMAL MODE - UPWARD DISPLACEMENT



REVERSIBLE PURGE MODE

A four-position damper reverses the fan path for 2-5 minutes.

Air sweeps laterally above the substrate, clears accumulated CO₂ and excess moisture, then returns to displacement mode.

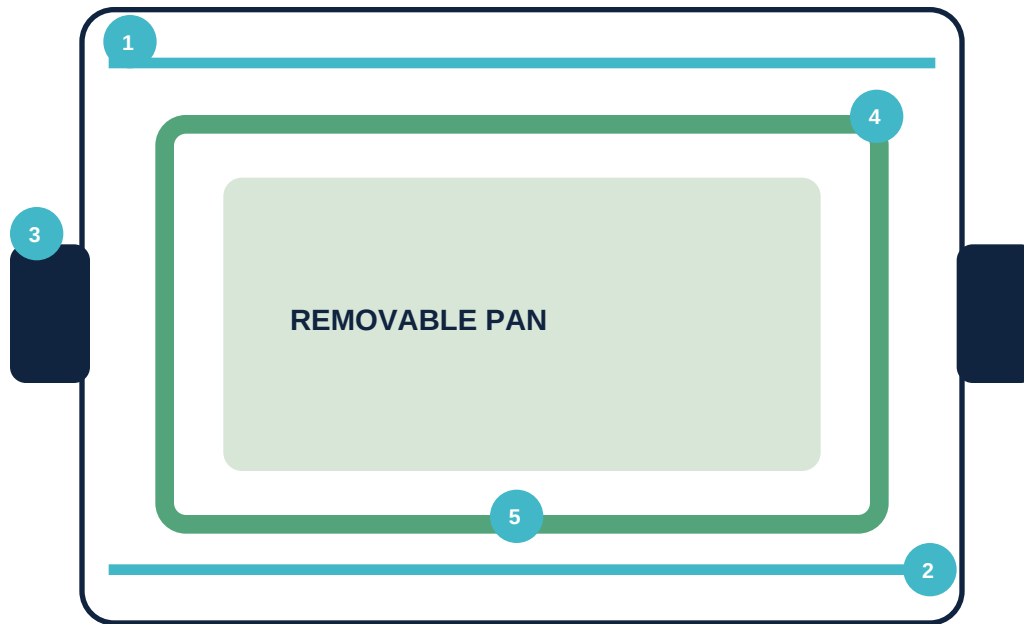
● Why slow delivery works

Large outlet area reduces local air speed and protects the moist boundary layer around developing fruits.

● Why tapering matters

Increasing perforation area with distance counteracts pressure loss and improves distribution uniformity.

CASSETTE ASSEMBLY



DESIGN DETAILS

- 1 Upper rail**
Prevents tilt during removal.
- 2 Lower rail**
Carries wet load.
- 3 Cool handles**
Stay outside chamber.
- 4 Label dock**
Batch ID and date.
- 5 Drain gap**
No pan-to-floor contact.

WHY THE CASSETTE MATTERS

- **Harvest ergonomics**
The crop slides toward the operator; no reaching around tubing, sensors or wet hardware.
- **Cleaning discipline**
The biological payload can be removed before service work, lowering accidental contact and splash risk.
- **Repeatability**
Identical cassette geometry makes pan position, air clearance and sensor height reproducible.
- **Future flexibility**
Alternate inserts can support one pan, two pans, blocks, bags or experimental chambers.

THREE-STAGE HUMIDITY PATH

1**Fog knockdown**

Humidifier output enters an expansion cup where large droplets settle.

2**Wick equalization**

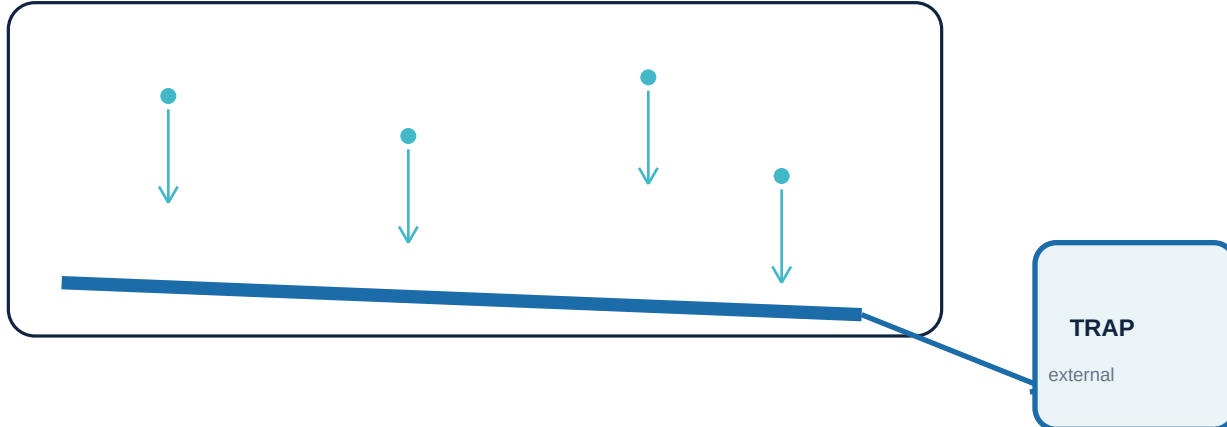
Air crosses a replaceable capillary panel that smooths output between cycles.

3**Ceiling-free delivery**

Humid air enters below the crop and rises without visible mist.

CONDENSATE RECOVERY LOOP

Walls and drip rails direct water to a pitched perimeter gutter.

**DESIGN INTENT**

The system controls relative humidity without treating airborne liquid water as the target.

That distinction reduces localized wetting, puddling and sensor saturation while preserving a high-humidity chamber.

07

Mechanical Controls & Failure Modes

Hardware should degrade gracefully before software intervenes

MANUAL CONTROL SURFACE

EXTERNAL SERVICE PANEL

AIR MODE

NORMAL / PURGE / PASSIVE

BLEED DIAL

1 2 3 4 5

FAN TRIM

LOW - HIGH

DRAIN VIEW

CLEAR TRAP

POWER

12 VDC

FAILURE RESPONSE

Power loss

Gravity flaps open

Passive FAE

Fan stall

Bypass port opens

No sealed chamber

Drain blockage

Sight trap rises

Visible warning

Humidifier stuck

Wick limits droplets

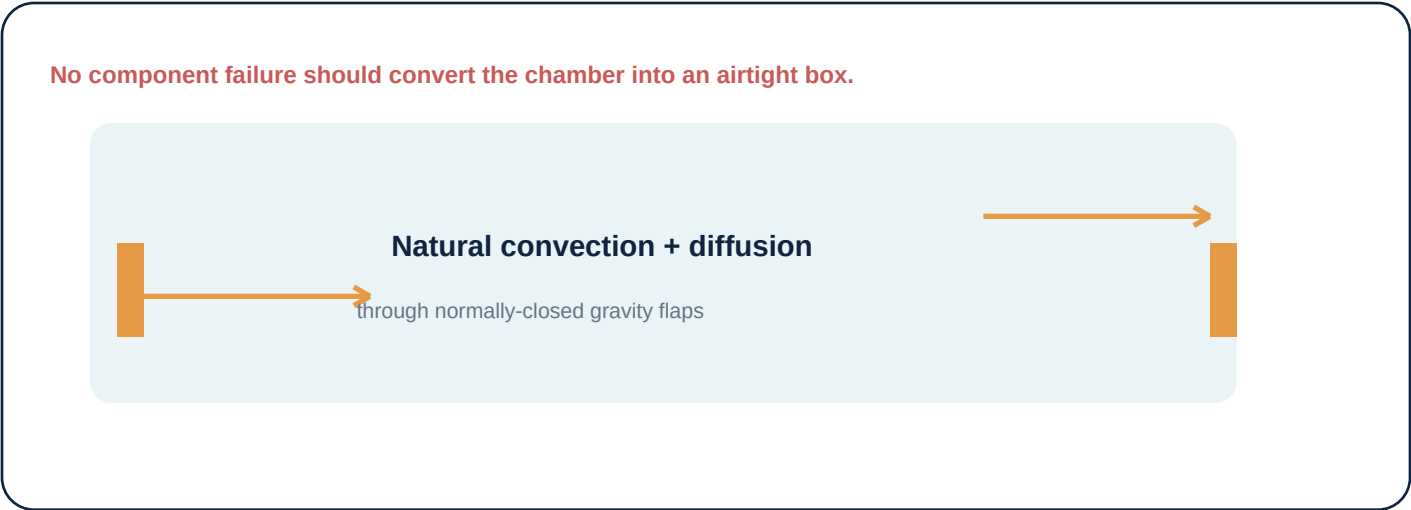
Reduced overshoot

Sensor fault

Manual dial remains

Operable offline

PASSIVE SURVIVAL PATH



08

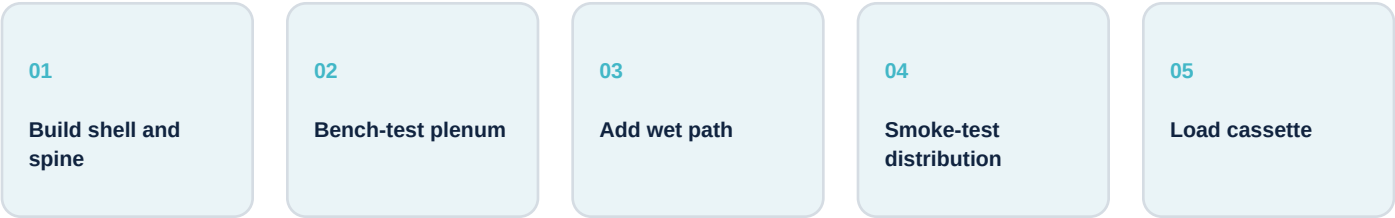
Prototype Build Specification

Hardware package for a first functional AX-80

BILL OF MATERIALS

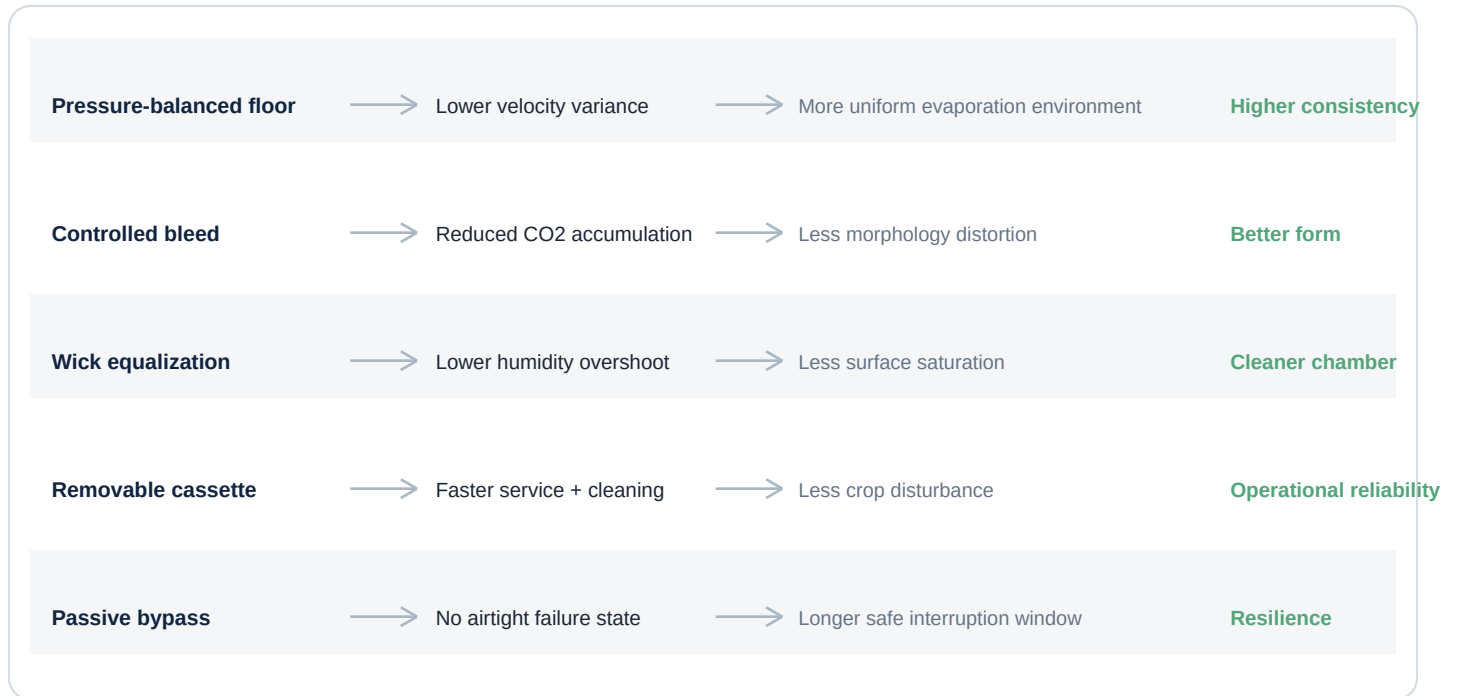
COMPONENT	QTY	FUNCTION
80 qt gasket tote	1	Structural shell
HDPE or polypropylene sheet	2	Plenum + cassette
12 V EC blower, speed controlled	1	Air mover
4-position rotary damper	1	Normal / purge / passive
Mist expansion cup + drain nipple	1	Droplet knockdown
Capillary wick cassette	1	Humidity equalizer
Silicone gravity flaps	2	Power-loss ventilation
Bulkhead fittings + clear tubing	set	Drain and humid air
External condensate trap	1	Visible recovery bottle
Stainless or polymer drawer rails	2	Grow cassette support

PROTOTYPE SEQUENCE



Critical prototype test: Use theatrical smoke or chilled vapor to verify even rise and absence of stagnant corners before biological use.

MECHANISM-TO-OUTCOME CHAIN



EVIDENCE BASE

Environmental control matters.

Controlled oyster-mushroom rooms have reported materially higher yield than ordinary rooms under the same seed batch; published work also identifies humidity and CO2 as direct growth variables.

CO2 distribution matters.

Research reviews report that elevated CO2 can suppress normal pileus differentiation. The AX-80 therefore prioritizes continuous low-rate removal plus a purge mode rather than humidity-only control.

Engineering conclusion

The architecture is strongly justified by transport and serviceability principles, but biological superiority must be validated by side-by-side trials.

10

Validation Plan & Decision

How to prove whether AX-80 earns the Dreamer name

A/B PROTOTYPE TRIAL

TEST DESIGN		PASS CRITERIA	
Control	Current direct-inlet Dreamer tub	RH spread	<= control
Prototype	AX-80 with same culture and substrate	CO2 peaks	lower
Replicates	Minimum 3 cycles per configuration	Visible mist	none
Hold constant	Culture, spawn ratio, substrate, light, temperature	Water use	not higher
Measure	RH variance, CO2, water use, yield, morphology, labor	Cleaning time	25% lower
		Yield	non-inferior

DECISION

BUILD THE AIRFRAME FIRST.

Do not add AI, cameras or automation until smoke testing and passive-failure testing pass.

SELECTED TECHNICAL REFERENCES

1.

Sakamoto, Y. (2018). Influences of environmental factors on fruiting body induction, development and maturation in mushroom-forming fungi. Fungal Biology Reviews.

2.

Guragain, D.P. et al. (2024). A low-cost centralized IoT ecosystem for enhancing oyster mushroom cultivation. Journal of Agriculture and Food Research, 15, 100952.

3.

Chong, J.L. et al. (2023). IoT-based environmental monitoring and control system for oyster mushroom cultivation. Biosensors, 13(1), 98.

4.

Penn State Extension (2024). Basic Procedures for Agaricus Mushroom Growing.

5.

Santos, A.F. et al. (2023). Eco-efficiency in mushroom production: HVAC equipment model. Applied Sciences, 13(10), 6129.

AX-80 3-CHAMBER PLATFORM

A scalable off-grid operating architecture combining three isolated chambers with shared power, humidity distribution and monitoring.

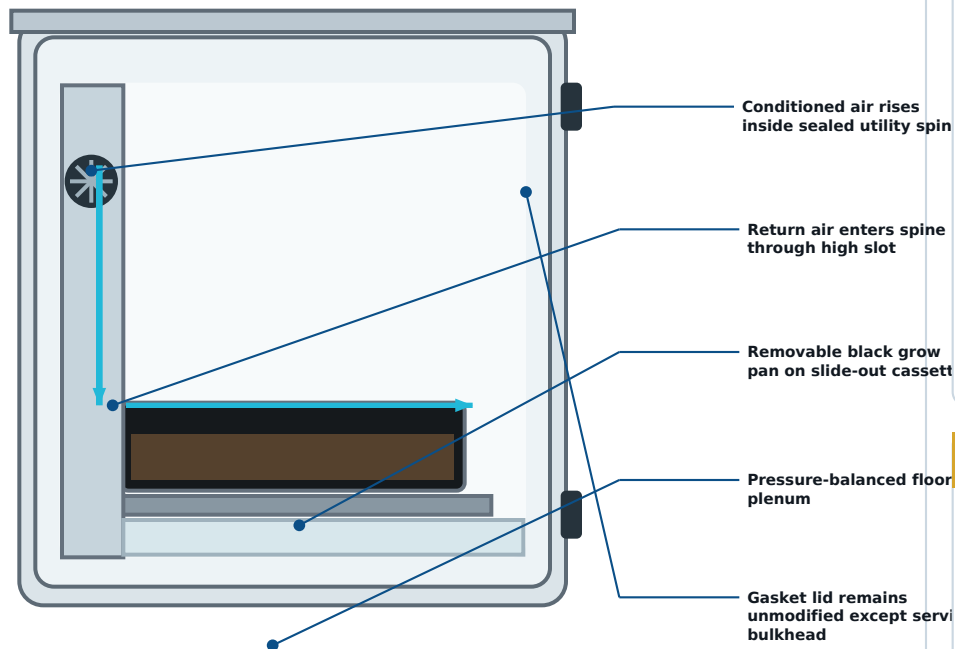
WHAT THIS APPROACH OPTIMIZES

- Modular throughput
- Independent branch isolation
- Rack-integrated utilities
- Low-voltage distribution
- Repeatable validation and scale-out

DREAMER TUB AX-80

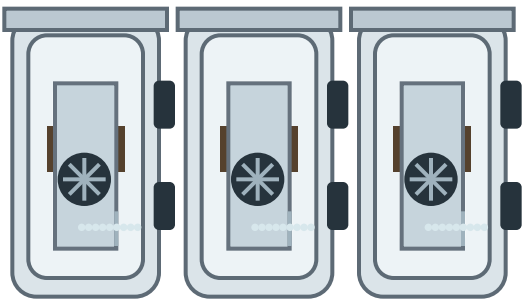
3-CHAMBER OFF-GRID FRUITING PLATFORM

1 SYSTEM ARCHITECTURE - ILLUSTRATED OVERVIEW



DESIGN INTENT Separate wet mechanical hardware from the biological zone while preserving easy access and passive fallback.

2 THREE CHAMBERS ON RACK

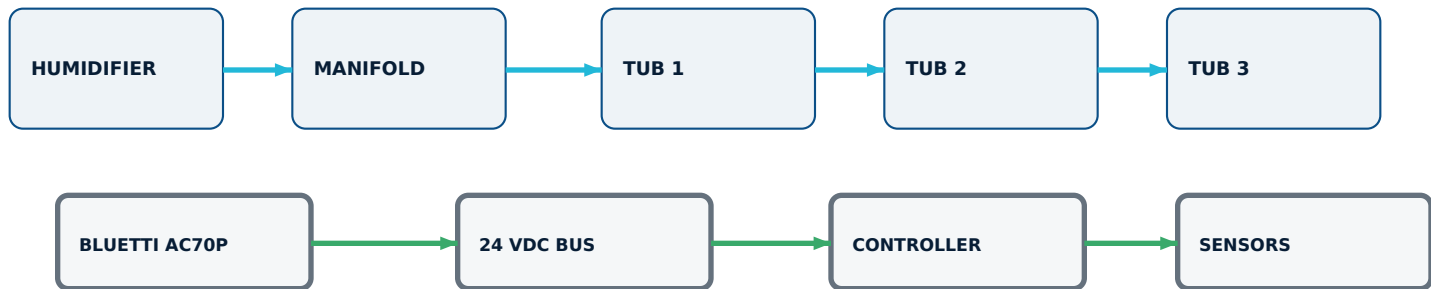


TUB 1 **TUB 2** **TUB 3**
Front-opening sideways orientation
Common humid-air manifold; isolated valves

3 CORE DIFFERENTIATORS

- Sealed rear service spine
- Tapered underfloor supply plenum
- Slide-out grow cassette
- Controlled recirculation + fresh-air bleed
- Gravity passive mode during outage

4 AIR + WATER + POWER TOPOLOGY



ONE HUMIDIFIER • THREE INDEPENDENT BALANCE VALVES • LOW-VOLTAGE INTERNAL HARDWARE • EXTERNAL CONDENSATE RETURN

5 PERFORMANCE HYPOTHESIS AND TESTABLE CLAIMS

UNIFORMITY

Reduced local wetting because air enters below the cassette through many small outlets.

STABILITY

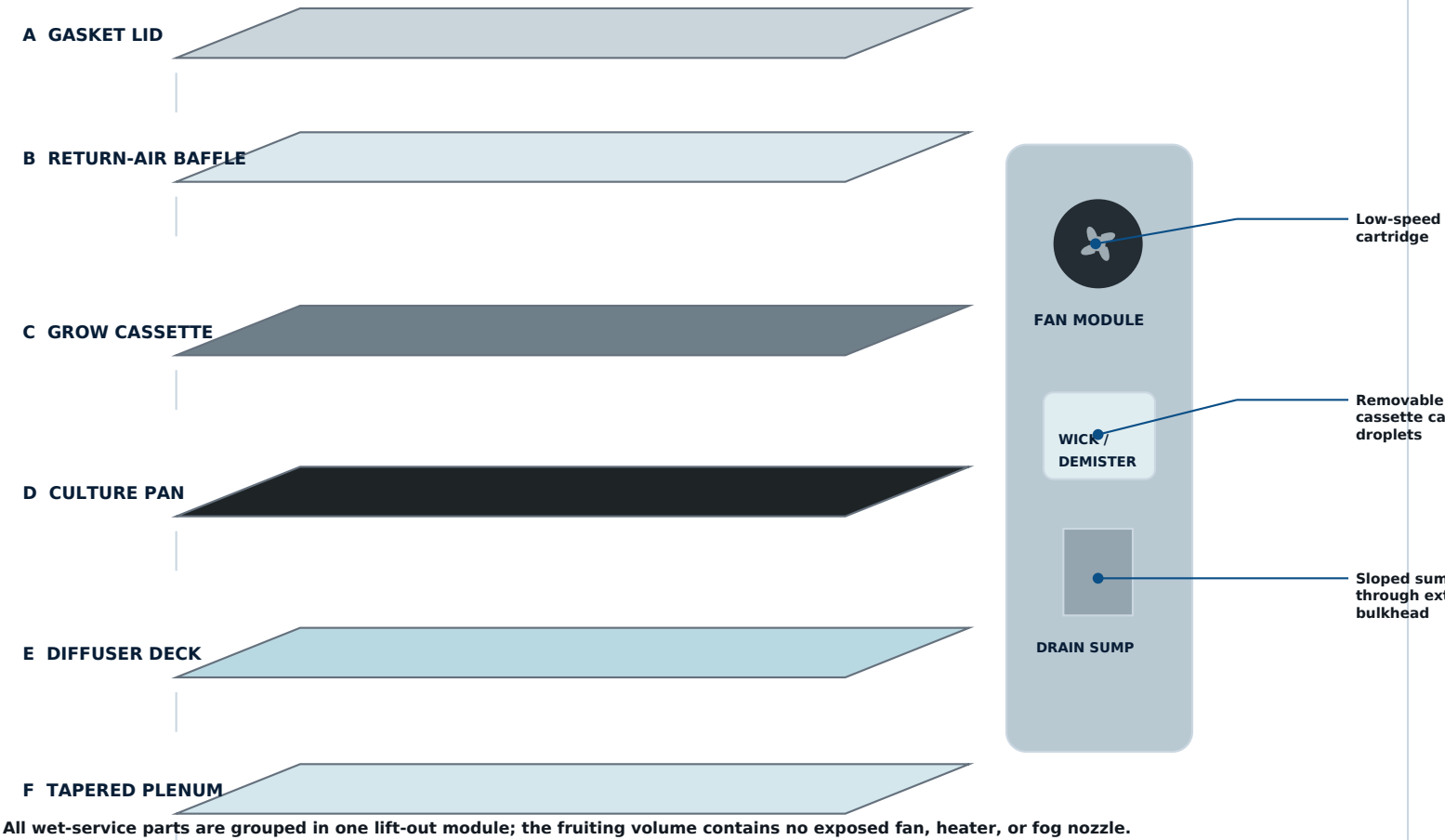
Recirculated air lowers RH oscillation and reduces unnecessary humidifier duty.

SERVICEABILITY

The cassette and service spine can be removed without disturbing fixed tub penetrations.

VALIDATION TARGETS <5% RH spread across nine points • no visible droplet strike • <10 min cassette removal • passive exchange after power loss

1 EXPLODED AXONOMETRIC - ONE CHAMBER



2 NUMBERED HARDWARE LIST

01 gasket-lid 80 qt tote	06 tapered supply plenum
02 HDPE spine shell	07 perforated diffuser deck
03 120 mm EC fan + guard	08 stainless cassette rails
04 coalescing mesh / wick tray	09 removable grow cassette
05 adjustable return damper	10 condensate bulkhead + trap

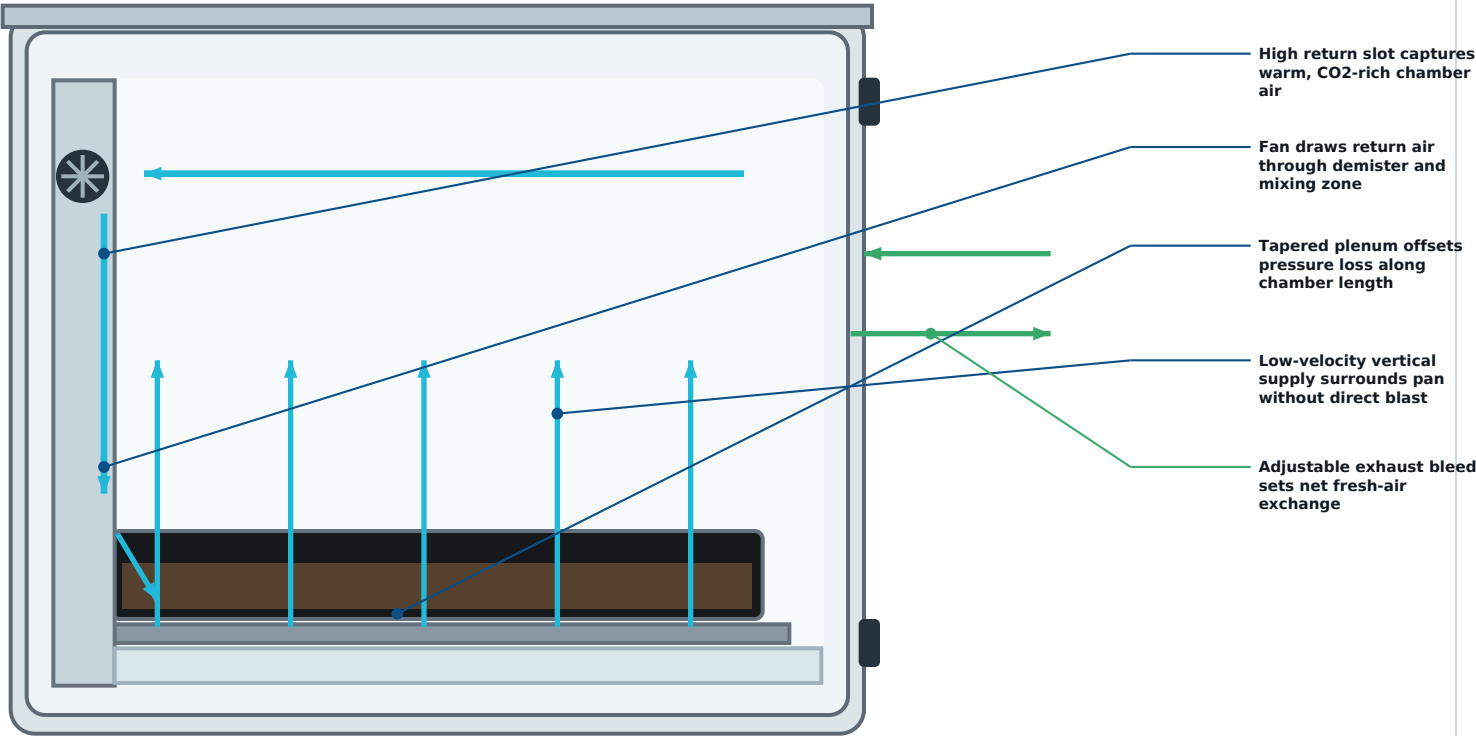
3 MATERIALS AND FABRICATION

STRUCTURAL	3 mm HDPE or polypropylene sheet
FASTENERS	Stainless M4, nylon locknuts, EPDM washers
SEALS	Closed-cell EPDM + removable silicone gaskets
DUCTS	Smooth-wall hose; no corrugated low points
CLEANING	Tool-less removal; radiused internal corners

4 ASSEMBLY DATUM AND TOLERANCE STRATEGY

PRIMARY DATUM	Existing tote floor and rear wall - no dependence on lid flatness.
CASSETTE CLEARANCE	6-8 mm each side to tolerate tote taper and molding variation.
PLENUM SEAL	Compression gasket only; diffuser deck lifts out for sanitation.
SERVICE MODULE	Single connector plate carries power, sensor, humid-air, and drain connections.

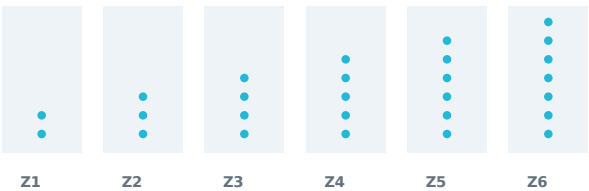
1 SECTIONAL AIRFLOW MAP



FLOW PRINCIPLE: recirculate most conditioned air, intentionally bleed a smaller fraction, and replace it through a filtered intake.

2 PRESSURE BALANCING LOGIC

DIFFUSER HOLE DENSITY



Fewer outlets near the fan; progressively greater open area toward the distal end.

3 TUNING CONTROLS

- FAN SPEED** Sets internal recirculation velocity.
- EXHAUST DIAL** Sets net air replacement rate.
- INTAKE FILTER** Adds controllable resistance and particulate capture.
- PLENUM SHIMS** Fine-tune left/right or near/far balance.
- CASSETTE HEIGHT** Adjusts canopy-to-return distance.

4 DESIGN CRITERIA AND PROTOTYPE MEASUREMENT GRID

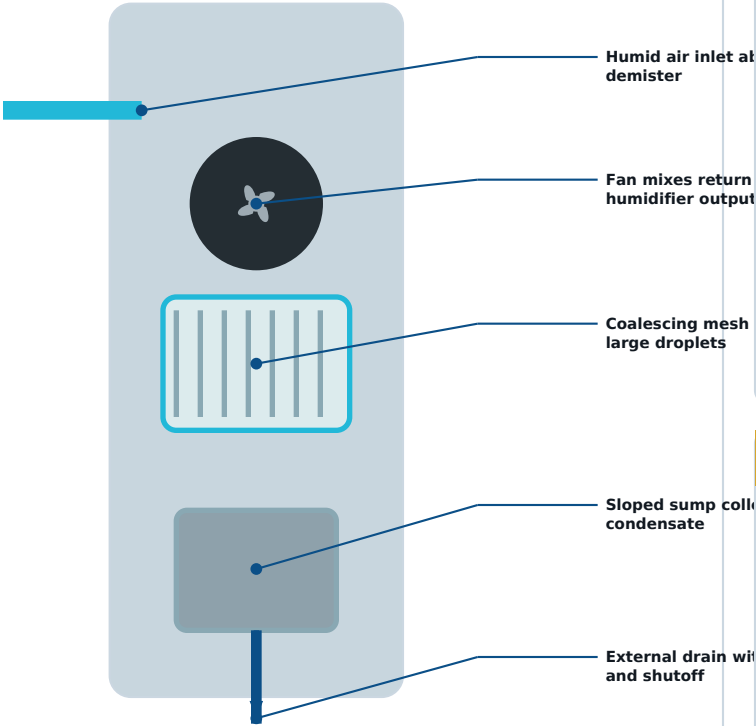
9-POINT RH / TEMP GRID



PASS CONDITIONS

- RH spread ≤ 5 percentage points at steady state
- No single-point air speed high enough to deform pins
- CO2 decay repeatable after defined pulse
- Visible fog absent in biological zone
- Recovery to RH setpoint after door opening

1 SERVICE SPINE WET PATH



The grow chamber receives saturated vapor-rich air, not a visible ultrasonic jet.

2 CONDENSATE RECOVERY ARCHITECTURE



Drain line slopes continuously outward; trap blocks uncontrolled air bypass.

Optional bottle scale quantifies water rejected by the chamber.

3 ANTI-PUDDLE GEOMETRY

WALL DRIP RAIL	Intercepts lid and wall runoff.
CASSETTE EAVE	Keeps condensate off substrate edge.
FLOOR SLOPE	Guides water toward service-side sump.
REMOVABLE LINER	Allows complete wipe-down outside tub.

4 MOISTURE STATES AND CORRECTIVE ACTIONS

DRYING RH below target; rising VPD ACTION Increase humid-air duty; reduce bleed; verify reservoir.	STABLE No visible fog; fine wall sheen ACTION Hold settings; monitor point spread.	OVER-WET Droplets on fruits or substrate ACTION Reduce humid-air pulse; increase demister area.	CONDENSING Cold wall / lid runoff ACTION Insulate inlet; reduce thermal swings; open drain.
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5 WHY THIS IS MORE EFFECTIVE THAN DIRECT-FOG INJECTION

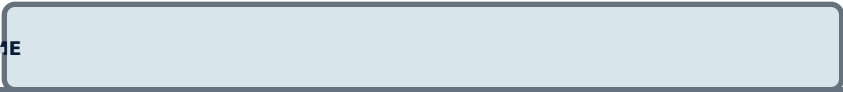
DIRECT INJECTION	High local droplet loading, wet spots, sensor contamination, and short humidity spikes.
CONDITIONED SPINE	Droplet separation + mixing + distributed delivery produces a slower and more spatially uniform moisture field.
TEST	Compare wall wetting, pan mass change, RH spread, and humidifier duty against a conventional side-port tub.

1 EXPLODED CASSETTE ASSEMBLY

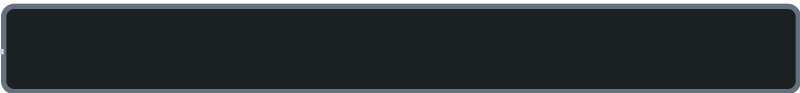
A FRONT HANDLE



B UPPER SPLASH FRAME



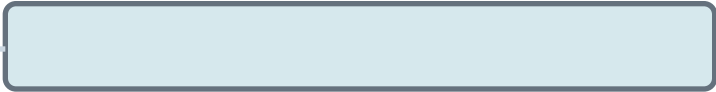
C CULTURE PAN



D SUPPORT CRADLE



E DRIP SHIELD



The biological payload becomes a removable appliance. All handling occurs at the cassette, not around fixed sensors or wet hardware.

2 FRONT ACCESS SEQUENCE

1. Pause fan and humid-air valve.
2. Release front gasketed access latch.
3. Pull cassette onto rack extension shelf.
4. Harvest or exchange pan.
5. Reinsert until rear magnetic stop engages.
6. Resume environment; verify recovery.

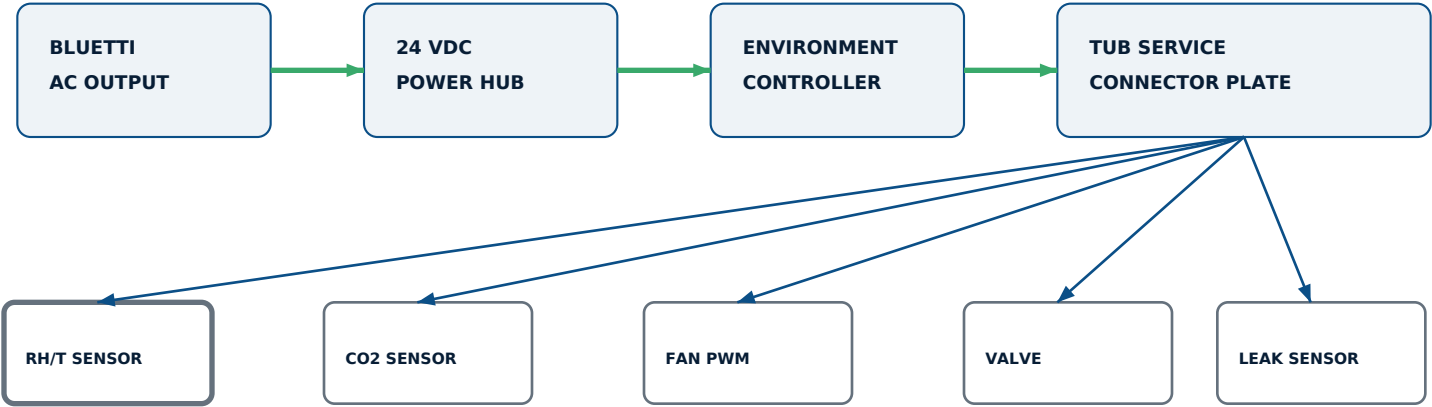
3 CASSETTE FEATURES

CAPTURED PAN	Pan cannot shift during insertion.
HEIGHT SHIMS	Set canopy clearance by species.
ID PLATE	Batch label visible through door.
DRIP EAVE	Deflects wall runoff outward.
AUTO-DOCK	Rear stop aligns sensor and air gap.

4 SERVICE-TIME COMPARISON

CONVENTIONAL TUB	Remove lid, reach around fans/tubing, lift pan vertically, wipe inaccessible bottom.
AX-80 CASSETTE	Front pull-out; no overhead lift; fixed hardware remains untouched; liner and cassette clean separately.
DESIGN TARGET	Full cassette removal in under 10 minutes with one person and no tool use.

1 CONTROL BLOCK DIAGRAM



All internal loads operate at low voltage. Mains power remains outside the humid environment.



12-PIN SERVICE PLATE

Fan • RH/T • CO2 • valve • leak • spare

2 SENSOR PLACEMENT

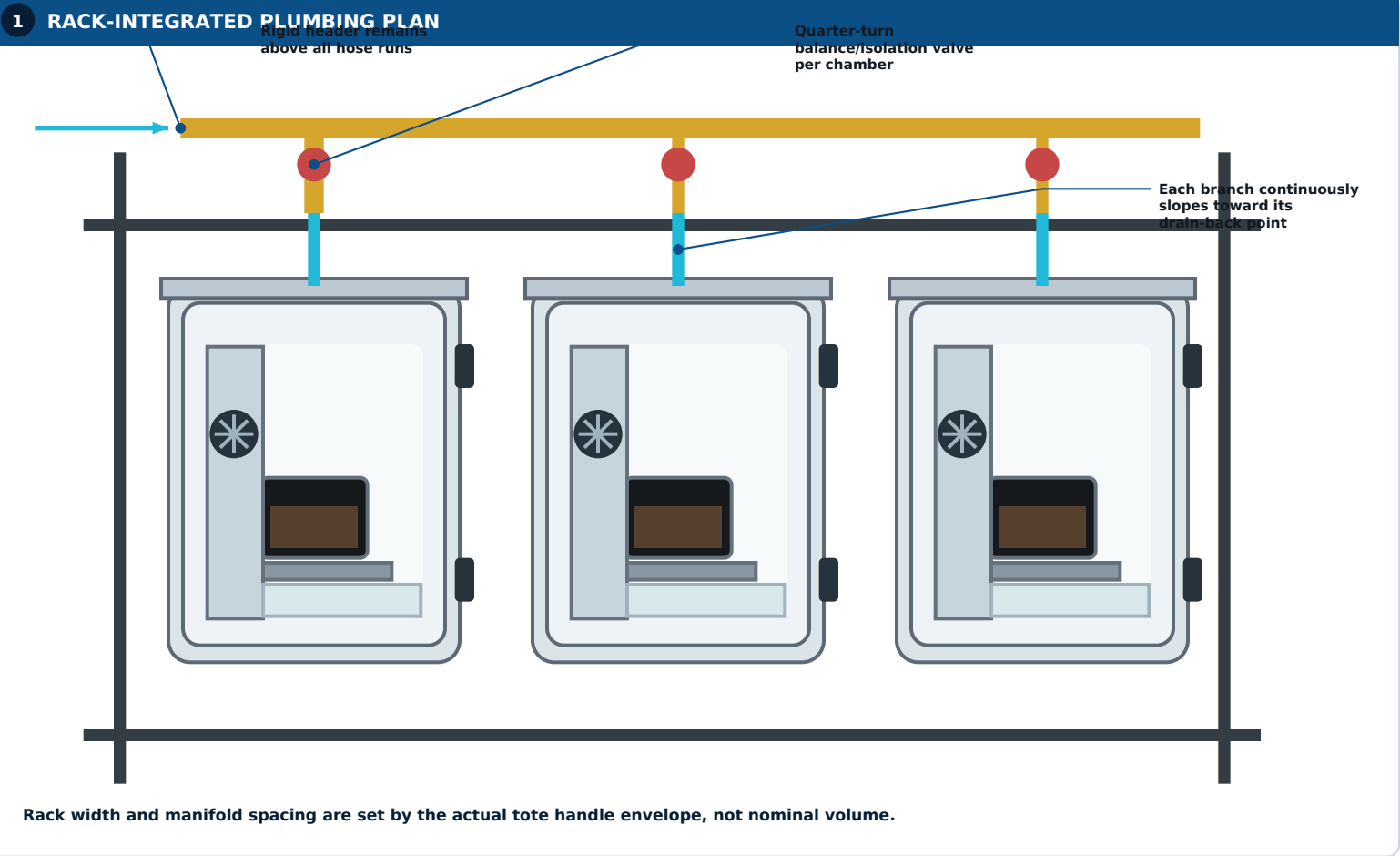
PRIMARY RH/T	Canopy height, shielded from direct supply.
SECONDARY RH/T	Inside spine after mixing stage.
CO2	Upper return zone; serviceable cartridge.
LEAK	Lowest point in service sump.
DOOR	Magnetic contact for recovery logging.

3 CONTROL HIERARCHY

1 SAFETY	Leak, overtemperature, blocked drain.
2 AIR	Continuous low fan; stage-based bleed.
3 HUMIDITY	Short duty pulses with minimum off-time.
4 ENERGY	Reduce duty before battery reserve is breached.
5 LOGGING	Store trends, access events, and water use.

4 ELECTRICAL SAFETY AND SERVICE RULES

- No exposed mains connections inside or beneath tubs.
- Use drip loops before every connector and controller.
- Fuse each tub branch independently.
- Use keyed connectors so fan and valve cannot be swapped.
- Mount battery and inverter outside the condensate fall zone.
- Design must remain passively breathable when all power is removed.



2

BALANCE PROCEDURE

- Open all three branch valves equally.
- Run humidifier and fans to steady state.
- Compare spine RH and chamber RH in each tub.
- Throttle the wettest branch in small increments.
- Repeat after adding/removing grow loads.
- Record valve index for each tub position.

3

RACK SERVICE FEATURES

PULL-OUT SHELF	Supports cassette during harvest.
DRIP PAN	Full rack-width emergency containment.
UTILITY RAIL	Carries manifold, wiring, and labels.
LEVELING FEET	Maintains drain slope on uneven floor.
FRONT CLEARANCE	No hoses cross the access face.

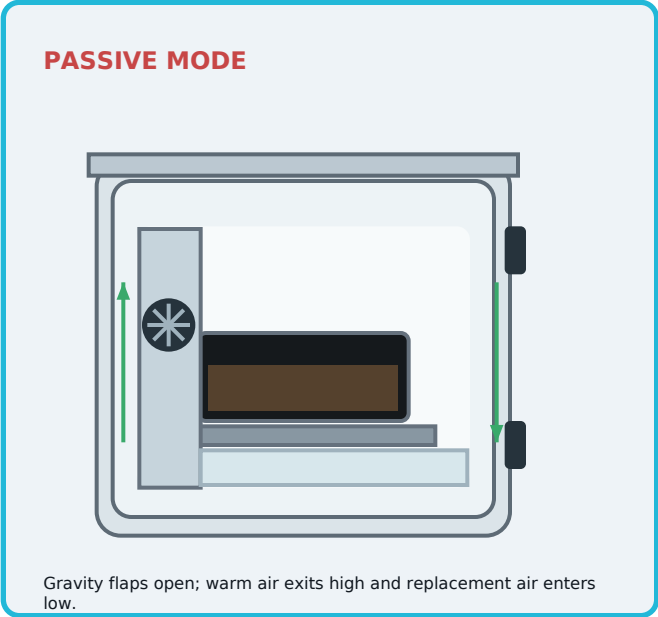
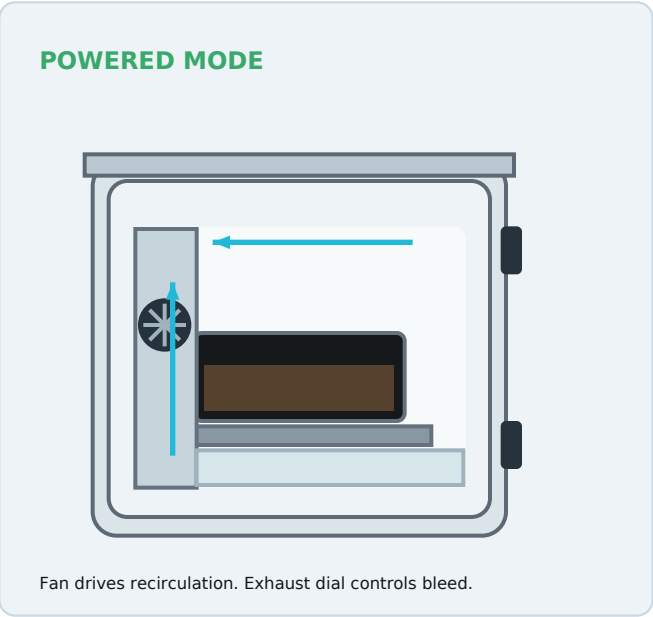
4

SINGLE-POINT FAILURES AND ISOLATION STRATEGY

HUMIDIFIER FAILS All tubs remain passively ventilated; local pans retain humidity.	ONE HOSE BLOCKS Only one branch affected; valve and quick-disconnect isolate it.	ONE TUB CONTAMINATES Close valve, unplug service connector, remove cassette and tub.	DRAIN BACKUP Leak sensor stops humidifier and flags service.	POWER LOSS Gravity flaps open and passive vents preserve exchange.
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1

POWERED MODE VERSUS PASSIVE MODE



The failure state is intentionally less optimized, but it remains breathable and observable.

2

FAILURE MODE AND EFFECTS ANALYSIS

FAILURE	DETECTION	AUTOMATIC RESPONSE	MANUAL RECOVERY
Fan stopped	RPM / current loss	Open passive flaps; stop fog	Replace fan cartridge
Drain blocked	Sump leak sensor	Stop humidifier	Clear trap and line
RH sensor wet	Rate-of-change mismatch	Use secondary sensor	Dry or replace probe
Valve stuck open	Branch RH divergence	Close manifold branch	Replace valve
Battery reserve low	State-of-charge threshold	Reduce duty; passive mode	Recharge / alternate power

3

RISK-REDUCTION DESIGN RULES

NO SINGLE WET ELECTRONIC

All controllers outside tub.

NO HIDDEN RESERVOIR

Every low point drains or is visible.

NO SEALED FAILURE

Passive vents bypass failed fan.

NO TOOL-LOCKED ACCESS

Cassette and fan cartridge remove by hand.

NO SHARED CONTAMINATION PATH

Each humid-air branch isolates.

NO DIRECT FOG ON SENSOR

Probe mounted outside supply plume.

1 PROTOTYPE COMPONENT MAP AND ESTIMATED QUANTITIES

80 qt gasket tote	3	Clear, front latches	Primary chamber shell
HDPE sheet	1	3 mm, approx. 24 x 48 in	Spines, plenums, diffusers
EC axial fans	3	120 mm, low-speed PWM	Internal recirculation
Humid-air hose	1	Smooth wall, 3/4-1 in ID	Distribution branches
Rigid manifold	1	Three valves + disconnects	Balancing / isolation
Mesh demister	3	Washable polypropylene	Droplet capture
Cassette rails	6	Stainless or polymer	Slide-out grow frame
Grow pans	-3	Removable, species dependent	Biological payload
Drain fittings	3	Bulkhead + trap + bottle	Condensate recovery
Sensors	-3	RH/T; optional CO2	Environment verification
24 VDC power hub	1	Fused branches	Low-voltage distribution
Rack	1	36-48 in wide, leveling feet	Structural support

2 BUILD SEQUENCE

1. Mock tote interior with cardboard templates.
2. Fit rear spine without cutting tub.
3. Build plenum and diffuser as one removable unit.
4. Fit cassette rails and verify front extraction.
5. Add one fan, one drain, and one humid-air inlet.
6. Instrument nine-point grid and validate one tub.
7. Only then replicate to tubs 2 and 3.

3 DO NOT OVERBUILD V1

INCLUDE	fan, plenum, cassette, drain, RH/T logging
DEFER	heater, camera, AI control, motorized dampers
Reason:	air and moisture distribution must be proven before automation adds complexity.
Prototype success is a stable climate map and easy service - not maximum feature count.	

4 ESTIMATED PROTOTYPE EFFORT

FABRICATION	COMMISSIONING	VALIDATION
1-2 weekends	2-4 days	One complete fruiting cycle
Cut, heat-bend, drill, gasket, and test-fit.	Map RH, tune valves, confirm drain behavior.	Compare against an unchanged control tub.

1

MECHANISM-TO-OUTCOME RATIONALE

DISTRIBUTED SUPPLY	Many low-velocity outlets reduce localized drying and direct droplet impact.	More uniform surface conditions across the pan.
RECIRCULATION + BLEED	Most conditioned air is retained while a controlled fraction is exchanged.	Lower humidity cycling with deliberate CO2 removal.
DEMISTING SPINE	Large droplets are removed before air reaches the grow chamber.	Less wall runoff, sensor wetting, and fruit-body spotting.
SLIDE-OUT CASSETTE	Biological load separates from fixed hardware.	Faster harvest, cleaning, and contaminated-batch isolation.
PASSIVE FALLBACK	Gravity vents provide an alternate air path when powered flow stops.	Reduced consequence of fan, controller, or battery failure.

These are engineering hypotheses. The design is justified by measurable mechanisms, not by claiming guaranteed biological yield.

2

A/B VALIDATION PROTOCOL

METRIC	CONTROL TUB	AX-80	PASS CRITERION
Nine-point RH spread	Record	Record	AX-80 lower spread
Visible droplet strike	Photo score	Photo score	None on fruiting surface
CO2 decay after pulse	Time constant	Time constant	Repeatable + tunable
Humidifier duty	Minutes/day	Minutes/day	Equal or lower
Water collected	g/day	g/day	Stable, drain-contained
Door-open recovery	Minutes	Minutes	Faster or comparable
Service time	Minutes	Minutes	AX-80 materially lower
Yield / quality	Mass + defects	Mass + defects	No degradation; trend positive

3

EVIDENCE BASE AND ENGINEERING CONCLUSION

ENVIRONMENTAL CONTROL	Published cultivation literature consistently treats humidity, temperature, air exchange, and CO2 as consequential fruiting variables.
AUTOMATION	Monitoring and closed-loop control are established approaches for maintaining mushroom-house conditions.
AX-80 CONTRIBUTION	The novelty is mechanical: condition, separate, and distribute the air before it reaches the biological zone, while making the grow load removable.
CONCLUSION	Proceed to a one-tub instrumented prototype. Scale to three chambers only after airflow uniformity, drainage, and passive fallback are demonstrated.

Selected sources: Sakamoto (2018), Mycoscience 59; Chong et al. (2023), Biosensors 13(1):98; Kavaliauskas et al. (2022), Applied Sciences 12:13040.

Decision lens	Integrated Cube	Adaptive Airframe	3-Chamber Platform
Primary objective	Finished autonomous product	Prove airflow and moisture mechanics	Scale independent grow capacity
Complexity	High	Medium	Medium-high
Best first build	Phase 1 tote prototype	One instrumented chamber	Only after one chamber passes
Strongest advantage	Integration and presentation	Uniformity and serviceability	Modularity and throughput
Main risk	Overbuilding before validation	Fabrication and tuning effort	Shared-utility imbalance
Recommended role	North-star product vision	Core R&D platform	Scale-out operating architecture

Recommended sequence: prove one airframe -> validate a live cycle -> replicate to three chambers -> package as the finished Cube.

ONE SYSTEM. THREE LEVELS OF AMBITION.

The Cube defines the destination.

The Airframe proves the mechanisms.

The 3-Chamber Platform proves the operating model.

BUILD THE AIRFRAME. MEASURE THE AIR. THEN SCALE.